

2024-10-22-Convexity

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Slide 1

Content

Convex Analysis [EVJ] Chapter 10

Def 1 $S \subseteq \mathbb{R}^m$ is convex if
for any $z_1, z_2 \in S \implies tz_1 + (1-t)z_2 \in S, \quad 0 \leq t \leq 1$

Description

This slide introduces the definitions of convex sets and convex combinations. **Definition 1** states that a set S in $\mathbb{R}^m$ is convex if for any two points z_1 and z_2 in S , the entire line segment connecting them is also contained within S . This is illustrated graphically by a line segment connecting two points in the set. **Definition 2** defines a convex combination of points $z_1, z_2, \dots, z_n$ in $\mathbb{R}^m$ as a weighted sum of these points, where the weights t_i are non-negative and sum to 1. This is illustrated by examples of convex combinations of three points in $\mathbb{R}^2$, showing how the convex combinations form a convex polygon. The context of the slide is Convex Analysis, a subject within Linear Programming, where convex sets play a crucial role in optimization problems.

Figures

Figure 1:
Definition of
a convex set
in $\mathbb{R}^m$. A line
segment con-
necting any two
points in the set
 S must also lie
entirely within
 S .

$$\text{minimize } \sum_{i=1}^n t_i z_i$$

Figure 2:
Examples of
convex com-
binations of
points z_1, z_2, z_3
in $\mathbb{R}^2$. The
shaded regions
illustrate the
convex combi-
nations of the
points, which
form a convex
polygon.

Slide 2

Content

Thm 10.1 S is convex $\iff$
 it contains all conv. comb. of points in S
pf $\iff$
 Suppose S contains all conv. comb. of pts in S
 Then clearly, for any $z_1, z_2 \in S$
 $tz_1 + (1-t)z_2 \in S$ (conv. comb. of z_1, z_2)
 $\implies S$ is convex

Description

This slide presents a portion of the proof for Theorem 10.1, which states that a set S is convex if and only if it contains all convex combinations of its points. The slide shows the "if" direction of the proof. It begins by assuming that S contains all convex combinations of its points. Then, it demonstrates that for any two points z_1 and z_2 in S , their convex combination $tz_1 + (1-t)z_2$ (where $0 \leq t \leq 1$) is also in S . This directly implies that S is convex, as per the definition of a convex set. The notation "conv. comb." stands for "convex combination". The context of the course (Linear Programming) is relevant because convex sets are fundamental to optimization problems in linear programming.

Figures

$$\sum_{i=1}^n \lambda_i x_i$$

Figure 3: A statement of Theorem 10.1, showing the equivalence between a set being convex and containing all convex combinations of its points. The proof is indicated by "pf".

Slide 3

Content

Thm 10.1 S is convex $\iff$ it contains all conv. comb. of points in S

pf " $\implies$ " Suppose S is convex.

$n = 2$: $z_1, z_2 \in S \implies t_1 z_1 + t_2 z_2 \in S, t_1, t_2 \geq 0, t_1 + t_2 = 1$

$n = 3$: $z_1, z_2, z_3 \in S \implies t_1 z_1 + t_2 z_2 + t_3 z_3 = (t_1 + t_2) \left(\frac{t_1}{t_1 + t_2} z_1 + \frac{t_2}{t_1 + t_2} z_2 \right) + t_3 z_3 \in S$

$t_1 + t_2 + t_3 = 1, t_1, t_2, t_3 \geq 0$

Description

This slide continues the proof of Theorem 10.1, showing that if a set S is convex, then it contains all convex combinations of its points. The proof proceeds by induction. The base case ($n = 2$) is shown: if z_1 and z_2 are in S , then any convex combination $t_1 z_1 + t_2 z_2$ (where $t_1, t_2 \geq 0$ and $t_1 + t_2 = 1$) is also in S by the definition of convexity. The inductive step ($n = 3$) is then demonstrated. It shows that if z_1, z_2, z_3 are in S , then a convex combination $t_1 z_1 + t_2 z_2 + t_3 z_3$ can be rewritten as a convex combination of a convex combination of z_1 and z_2 and z_3 . Since the inner convex combination is in S (by the base case), and the outer convex combination is also in S (by the definition of convexity), the entire expression is in S . The notation $t_1, t_2, t_3 \geq 0$ and $t_1 + t_2 + t_3 = 1$ indicates the non-negative weights and their sum to 1, which are crucial for convex combinations. The context of linear programming is relevant because convex sets are fundamental to optimization problems in linear programming.

Figures

to 2 ∇

Figure 4: A statement of Theorem 10.1, showing the equivalence between a set being convex and containing all convex combinations of its points. The proof is indicated by "pf".

$$(4.2) \quad \begin{aligned} & \text{maximize} \quad \sum_{j=1}^n c_j x_j \\ & \text{subject to} \quad \sum_{j=1}^n a_{ij} x_j = b_i \end{aligned}$$

Figure 5: A portion of the proof for the case $n=2$ and $n=3$.